

An Exact Boundary and Skin Atlas for the Finite Hatano–Nelson Chain

Route-A candidate HCS-C308

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Abstract

We solve the finite asymmetric nearest-neighbor chain across its full hopping cone. Positive open-boundary hopping is diagonally similar to a symmetric path matrix, yielding the exact Chebyshev spectrum. We also obtain a canonical biorthogonal sine basis, distinguish its asymmetry-free density from the right-amplitude skin envelope, and give exact conditioning, propagation, and resolvent formulas. Periodic Fourier spectra and every Hermitian, one-sided, zero, orientation-reversed, and small-size face complete the atlas. The singular open/periodic split is proved without a topology, disorder, or arithmetic claim.

1 Model and the algebraic core

For $N \geq 2$, let the open-boundary matrix $H_N = H_N(t_R, t_L)$ have $(H_N)_{j,j+1} = t_R$ and $(H_N)_{j+1,j} = t_L$, where $t_R, t_L \geq 0$. Evolution means $i\psi = H_N\psi$. When both hoppings are positive, put

$$g = \sqrt{t_R t_L}, \quad q = \sqrt{t_L/t_R}, \quad D = \text{diag}(1, q, \dots, q^{N-1}),$$

and let A_N be path adjacency. Define

$$\theta_m = \frac{m\pi}{N+1}, \quad S_{jm} = \sqrt{\frac{2}{N+1}} \sin(j\theta_m), \quad 1 \leq j, m \leq N.$$

Theorem 1 (finite-chain atlas). *For $t_R, t_L > 0$,*

$$D^{-1}H_N D = gA_N, \quad \det(zI - H_N) = g^N U_N\left(\frac{z}{2g}\right), \quad E_m = 2g \cos \theta_m. \quad (1)$$

The E_m are real and simple. In the canonical sine gauge,

$$R = DS, \quad L^T = S^T D^{-1}, \quad L^T R = I, \quad \kappa_2(R) = \max(q, q^{-1})^{N-1}. \quad (2)$$

Moreover,

$$e^{-itH_N} = DS \text{diag}(e^{-itE_m}) S^T D^{-1}, \quad (zI - H_N)^{-1} = D(zI - gA_N)^{-1} D^{-1}. \quad (3)$$

For $x = z/(2g)$ and $z \notin \sigma(H_N)$,

$$[(zI - H_N)^{-1}]_{ij} = \frac{q^{i-j} U_{\min(i,j)-1}(x) U_{N-\max(i,j)}(x)}{g U_N(x)}. \quad (4)$$

For the oriented ring with $N \geq 3$, let $(Cv)_j = v_{j+1 \bmod N}$ and $H_N^{\text{per}} = t_R C + t_L C^{-1}$. It is normal and its Fourier spectrum is

$$E_m^{\text{per}} = t_R e^{ik_m} + t_L e^{-ik_m}, \quad k_m = 2\pi m/N. \quad (5)$$

Thus $\Re E_m^{\text{per}} = (t_R + t_L) \cos k_m$ and $\Im E_m^{\text{per}} = (t_R - t_L) \sin k_m$.

On $t_R = t_L > 0$ both matrices are Hermitian. If exactly one hopping is positive, OBC is one nilpotent N -Jordan block, while PBC is a unitarily diagonalizable scaled cyclic shift. If both vanish, the matrix is zero with eigenspace dimension N . Swapping t_R, t_L transposes OBC, reverses its right-amplitude edge, and complex-conjugates the PBC ellipse.

Proof. Since $D_j^{-1}t_R D_{j+1} = t_R q = g$ and $D_{j+1}^{-1}t_L D_j = t_L/q = g$, diagonal conjugation gives the first identity. Laplace expansion produces the continuant

$$P_0 = 1, \quad P_1 = z, \quad P_N = zP_{N-1} - t_R t_L P_{N-2}.$$

The recurrence for U_N gives (1). The discrete sine transform is orthogonal and diagonalizes A_N , which proves simplicity and reality. Conjugating its columns gives R , while $S^T D^{-1}$ is the exact left dual. Right multiplication by orthogonal S preserves singular values, so the extreme diagonal entries of D give (2). This equality concerns the stated sine gauge; arbitrary column rescaling changes an eigenvector-matrix condition number. Functional calculus gives (3). The standard continuant formula for the inverse of $zI - gA_N$, followed by diagonal conjugation, gives (4). Consequently, for real t ,

$$\|e^{-itH_N}\|_2 \leq \kappa_2(D), \quad \|(zI - H_N)^{-1}\|_2 \leq \frac{\kappa_2(D)}{\text{dist}(z, \sigma(H_N))}. \quad (6)$$

These are bounds, not asserted equalities. The unitary Fourier basis diagonalizes C , proving (5), the ellipse parametrization, and normality. On a one-sided OBC axis the surviving shift satisfies $H^N = 0$, $H^{N-1} \neq 0$, and $\text{rank}(H^k) = N - k$; hence there is one Jordan chain of length N . On the ring, $C^N = I$ and its Fourier eigenvalues are distinct. Direct substitution proves the Hermitian, zero, and orientation statements. $\square$

2 What is, and is not, skin localization

The right and left components in the canonical dual normalization are

$$R_{jm} = q^{j-1} S_{jm}, \quad L_{mj}^T = q^{-(j-1)} S_{jm}.$$

Thus ordinary right amplitudes carry an exponential envelope toward the left for $q < 1$ and toward the right for $q > 1$. Pointwise biorthogonal density is a different object:

$$L_{mj}^T R_{jm} = S_{jm}^2,$$

which is independent of q and sums to one. We use “skin” only for the first, amplitude-level statement.

3 Boundary faces and a singular limit

At fixed N , taking $t_L \downarrow 0$ with $t_R > 0$ collapses every positive-OBC eigenvalue to zero and sends the canonical conditioning in (2) to infinity. The limit is defective. Under PBC, the same limit has eigenvalues $t_R e^{ikm}$ and keeps its Fourier eigenbasis. The transpose statement holds when $t_R \downarrow 0$. This proves a boundary-condition-sensitive singular limit; it does not prove a winding invariant or a protected edge mode.

The one-sided OBC formulas also close dynamically:

$$e^{-itH} = \sum_{k=0}^{N-1} \frac{(-itH)^k}{k!}, \quad (zI - H)^{-1} = \sum_{k=0}^{N-1} \frac{H^k}{z^{k+1}} \quad (z \neq 0).$$

For $N = 2$, the OBC theorem remains unchanged. A two-site cyclic shift obeys $C = C^{-1}$, so the two oriented neighbors coincide and $H_2^{\text{per}} = (t_R + t_L)C$. We therefore state the standard oriented-ring ellipse theorem for $N \geq 3$ and record $N = 2$ only as this explicit convention face. At finite N , (5) gives finitely many points on an ellipse, not the entire ellipse.

4 Executable evidence and scope

The analytic proof covers all stated N and parameter faces. Finite evidence is regression only: 40 positive-OBC continuants and condition numbers, 21 resolvent determinant/log-derivative traces, 18 one-sided Jordan/PBC cases, 36 PBC power-trace cases, and eight semantic boundary faces (123 rows total). An independent checker recomputes every cell with strict JSON/YAML types; a separate symbolic lane checks the identities; isolated double replay demands byte equality; and repaired-hash mutations attack formulas, owners, boundaries, types, lists, and parsers.

The Route-A tuple is (A0_FAIL, A1_FAIL, A2_FAIL, A3_FAIL, A4_FORMAL_HINT) and the overall verdict is ROUTE_A_REJECTED; Route B is locked. A finite Chebyshev determinant is not an Euler factor, the finite spectrum is not a target zero set, and a nonunitary similarity to a source-local symmetric matrix is not a same-clock self-adjoint Hilbert–Pólya construction. We claim no disorder localization, interactions, topological invariant, automorphy, root number, or literature priority. The frozen scope is NO_BAD_EULER_OR_ROOT_NUMBER.

AI-use disclosure. An AI system assisted with algebraic organization, code generation, and prose. Every released formula is supported by the displayed proof and independent executable checks; source ownership and claim boundaries were reviewed explicitly. No external review or authorship is represented.

References

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